

30-Day Readmission of Diabetic Patients

SQL, exploratory analysis, statistical inference and a risk model on 69,970 patients from 130 US hospitals

01 Data audit

02 SQL

03 EDA

04 Inference

05 Risk model

06 Power BI

69,970

patients, one encounter each

8.97%

readmitted within 30 days

18.4%

had HbA1c measured

0.659

test ROC-AUC, XGBoost

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Data: UCI Diabetes 130-US Hospitals, 1999 to 2008
Python, DuckDB, statsmodels, scikit-learn, XGBoost, SHAP, Power BI

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Executive summary

About 1 in 11 diabetic patients in this data came back to hospital within 30 days of discharge. This report looks at who those patients are, whether measuring HbA1c during the stay is linked to fewer readmissions, and how well readmission can be predicted at the point of discharge.

The data is the UCI Diabetes 130-US Hospitals dataset: 101,766 inpatient encounters from 1999 to 2008. After keeping each patient's first encounter and removing discharges to hospice or death, **69,970 patients** remain. The work runs through seven notebooks: data download, audit, SQL business questions in DuckDB, exploratory analysis, statistical inference, a risk model, and an export for the Power BI dashboard.

8.97%

readmitted within 30 days

3.6×

adjusted odds with 3+
prior inpatient stays

18.4%

of patients had HbA1c
measured

0.659

test ROC-AUC of the
XGBoost risk model

Main findings

- 1. Prior hospital use is the strongest risk factor.** Patients with three or more inpatient stays in the previous year are readmitted at 26.5%, against 8.1% for patients with none. After adjusting for everything else, their odds are 3.61 times higher (95% CI 3.09 to 4.23).
- 2. Where the patient goes after discharge matters almost as much.** Patients transferred to another care facility come back at 15.2%, patients sent home at 7.0% (adjusted OR 2.17). Discharge destination is also the top driver in the model's SHAP values.
- 3. HbA1c testing is rare, and its effect depends on the diagnosis.** Only 18.4% of patients were tested, and only 35.9% of those admitted for diabetes. Across all patients testing is not linked to readmission once patient mix is adjusted for (adjusted OR 0.96, propensity matched OR 0.99, both intervals include 1). But the effect differs by diagnosis ($p = 0.0067$), and testing goes with lower odds in injury (OR 0.61), respiratory (0.76) and diabetes (0.81) admissions.
- 4. Readmission can be predicted moderately well.** The XGBoost model reaches a test ROC-AUC of 0.659 and a PR-AUC of 0.183, twice the random baseline, with close calibration. The top 20% of patients by predicted risk hold 38% of readmissions. At a cost-based threshold of 0.09 the model flags 36% of patients and catches 56% of readmissions.

What it means in practice

A care team can use the risk score to order its follow-up list, and patients with prior inpatient stays or a transfer to another facility are the obvious first group for a structured discharge and an early call. HbA1c testing in injury, respiratory and diabetes admissions is worth a closer look. These are associations from observational data from 1999 to 2008, and the model's accuracy is limited by what the data leaves out, such as social support and follow-up care (Section 8).

1 Introduction

What this project asks, what data it uses, and how the work is split across seven notebooks.

1.1 Background

A readmission within 30 days of discharge is one of the standard quality measures for hospitals in the United States. It costs the hospital and the patient, and it often points to something that went wrong in the discharge plan or the follow-up. Since 2012 the Hospital Readmissions Reduction Program has cut Medicare payments to hospitals with higher than expected readmission rates for selected conditions, so hospitals have a direct reason to find the patients most likely to come back.

Diabetic patients are a large share of inpatients, and they are admitted for all kinds of reasons, not only for their diabetes. Strack et al. (2014) used the same dataset as this project to ask whether measuring HbA1c during the hospital stay is linked to readmission. They found that it was, and that the relationship depended on the primary diagnosis. That question is re-examined in Section 5 with three different methods.

1.2 Questions

1. **Where is readmission concentrated?** By age, diagnosis, discharge destination, prior hospital use, length of stay and medication (Sections 3 and 4).
2. **Is HbA1c testing linked to lower readmission once patient mix is accounted for**, and does that depend on why the patient was admitted? (Section 5)
3. **How well can 30-day readmission be predicted at discharge**, and what drives the prediction? (Section 6)
4. **How would a care team use the result?** Risk tiers, a decision threshold and a what-if scenario for the Power BI dashboard (Sections 6 and 7).

1.3 Data

Table 1.1 Dataset summary.

Item	Detail
Source	UCI Machine Learning Repository, "Diabetes 130-US Hospitals for Years 1999 to 2008"
Coverage	130 US hospitals and integrated delivery networks, 1999 to 2008
Raw size	101,766 inpatient encounters, 71,518 unique patients, 50 columns
Inclusion (by the data owners)	Diabetic encounter, stay of 1 to 14 days, lab tests and medications given
Target	<code>readmitted</code> : <30 days, >30 days, or no record of readmission
Analysis cohort	69,970 patients, one encounter each (Section 2)

1.4 Workflow

The project is split into seven Google Colab notebooks that run in order. Each one reads from and writes to a fixed folder layout on Google Drive, so any notebook can be rerun on its own once the earlier ones have produced their files. SQL work is done in DuckDB directly on Parquet files. Statistics use statsmodels and SciPy, the model uses scikit-learn, XGBoost and SHAP, and the dashboard is built in Power BI.

Table 1.2 Notebooks and what each one produces.

Notebook	Name	What it does
01	Data download	Downloads the UCI files and converts them to Parquet
02	Data audit	Checks keys, lookups, missing values; applies cohort rules
03	SQL business questions	13 questions in DuckDB SQL, one CSV per answer
04	Exploratory analysis	Effect sizes, segment rates, HbA1c testing gap
05	Statistical inference	Logistic regression, interaction test, propensity score matching
06	Readmission risk model	Logistic regression vs XGBoost, threshold, SHAP, risk tiers
07	Power BI export	Star schema tables and a what-if scenario for the dashboard

2 Data preparation and cohort

Notebooks 01 and 02. Loading the raw files, auditing them, and building one clean row per patient.

2.1 Loading

The raw CSV marks missing values with `?`. It is read as NULL during the Parquet conversion so missingness shows up as ordinary nulls later. The three diagnosis columns are forced to text because ICD-9 codes mix numbers with `V` and `E` codes, and a numeric type would silently drop those. The ID mapping file holds three lookup tables stacked in one CSV, so it is split into three Parquet files.

Code 2.1 Converting the raw CSV to Parquet with DuckDB and splitting the lookup file.

01_download_data.ipynb, cell 5

```
con = duckdb.connect()
diag = {c: 'VARCHAR' for c in ('diag_1', 'diag_2', 'diag_3')}
con.execute(f"COPY (SELECT * FROM read_csv('{RAW}/diabetic_data.csv', header=true, nullstr='?', sample_size=-1,
auto_type_candidates=['BIGINT', 'DOUBLE', 'VARCHAR'], types={diag})) TO '{PQ}/diabetic_data.parquet' (FORMAT
PARQUET)")
mp = next(f for f in RAW.glob('*.csv') if 'mapping' in f.name.lower())
for part in filter(str.strip, re.split(r'\n\s*?\s*\n', mp.read_text().replace('\r', ''))):
    m = pd.read_csv(io.StringIO(part.strip()), skipinitialspace=True); m.to_parquet(PQ/f"
{m.columns[0].removesuffix('_id')}.parquet", index=False)
print(*sorted(f.name for f in PQ.glob('*.parquet')), sep='\n')
```

2.2 Audit

`encounter_id` is unique across all 101,766 rows. Patients are not: 23.5% of the 71,518 patients have two or more encounters, and 1,590 have five or more. Repeat visits from the same person are not independent observations, which matters for every test and model later on.

Every admission type, discharge and admission source ID matches its lookup table, but some lookups are empty placeholders. 10,396 encounters have an admission type described as `NULL`, `Not Available` or `Not Mapped`, and the same happens for 4,680 discharge codes and 7,067 admission source codes. These are hidden missing values and become `Unknown` in the clean table.

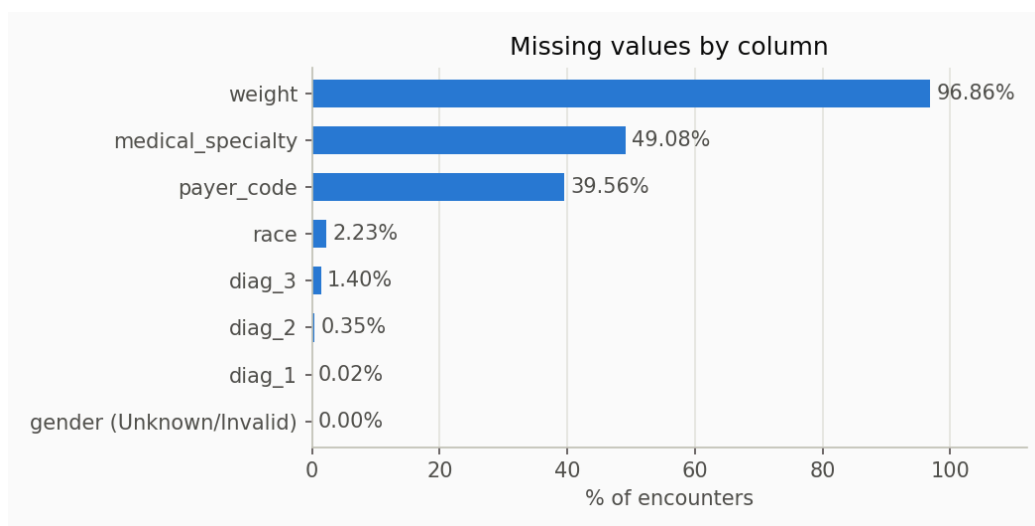


Figure 2.1 Missing values by column in the raw data (101,766 encounters).

Table 2.1 Missing value decisions.

Column	Missing	Decision
weight	96.9%	Dropped
medical_specialty	49.1%	Kept, missing becomes Unknown
payer_code	39.6%	Kept, missing becomes Unknown
race	2.2%	Kept, missing becomes Unknown
diag_1 to diag_3	0.02% to 1.4%	Kept, missing becomes its own diagnosis group
gender = Unknown/Invalid	3 rows	Rows removed
A1Cresult, max_glu_serum = None	83% and 95%	Recoded to Not Tested (the test was not done, it is not missing data)
citoglipton, examide	never prescribed	Dropped (constant columns)

2.3 Cohort rules

Three rules turn encounters into an analysis cohort. First, the three encounters with invalid gender are removed. Second, only the first encounter of each patient is kept, so every row is an independent patient. The data has no admission dates, so the lowest `encounter_id` stands in for the first visit. Third, patients discharged to hospice or who died in hospital are removed, since they cannot be readmitted. These rules follow the same idea as Strack et al. (2014).

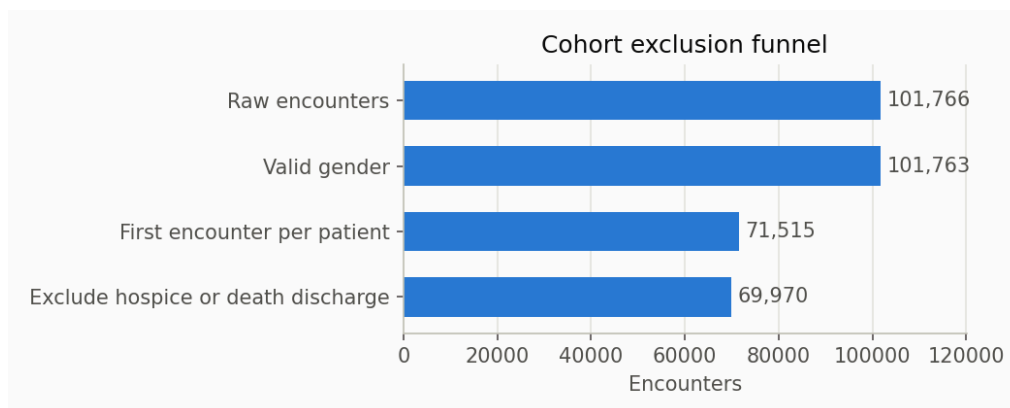
Code 2.2 Cohort rules and the row count after each step.

02_data_audit.ipynb, cell 16

```
con.execute("CREATE OR REPLACE TABLE base AS SELECT *, ROW_NUMBER() OVER (PARTITION BY patient_nbr ORDER BY
encounter_id) rn FROM diabetic_data WHERE gender <> 'Unknown/Invalid'")
EXCL = '(11, 13, 14, 19, 20, 21)'
steps = [('Raw encounters', 'diabetic_data', 'TRUE'), ('Valid gender', 'base', 'TRUE'), ('First encounter per
patient', 'base', 'rn = 1'),
('Exclude hospice or death discharge', 'base', f'rn = 1 AND discharge_disposition_id NOT IN {EXCL}')]
audit = pd.DataFrame([(s, con.sql(f"SELECT COUNT(*) FROM {t} WHERE {w}").fetchone()[0]) for s, t, w in steps],
columns=['step', 'n_rows']).assign(removed=lambda d: (-d.n_rows.diff()).fillna(0).astype(int), pct_kept=lambda
d: (100*d.n_rows/d.n_rows[0]).round(1))
```

Table 2.2 Rows kept at each cohort step.

step	n_rows	removed	pct_kept
Raw encounters	101,766	0	100.0
Valid gender	101,763	3	100.0
First encounter per patient	71,515	30,248	70.3
Exclude hospice or death discharge	69,970	1,545	68.8

**Figure 2.2** Cohort exclusion funnel. 68.8% of the raw encounters remain.

2.4 Diagnosis groups and the clean table

The three diagnosis columns hold over 700 distinct ICD-9 codes each. They are grouped into nine clinical categories with a DuckDB macro, using the code ranges from Strack et al. (2014). The macro is tested on a handful of known codes before it is applied.

Code 2.3 ICD-9 grouping macro.

02_data_audit.ipynb, cell 19

```
con.execute("""CREATE OR REPLACE MACRO icd_group(c) AS CASE
WHEN c IS NULL THEN 'Missing'
WHEN c LIKE 'V%' OR c LIKE 'E%' THEN 'Other'
WHEN c LIKE '250%' THEN 'Diabetes'
WHEN TRY_CAST(c AS DOUBLE) BETWEEN 390 AND 459.99 OR c LIKE '785%' THEN 'Circulatory'
WHEN TRY_CAST(c AS DOUBLE) BETWEEN 460 AND 519.99 OR c LIKE '786%' THEN 'Respiratory'
WHEN TRY_CAST(c AS DOUBLE) BETWEEN 520 AND 579.99 OR c LIKE '787%' THEN 'Digestive'
WHEN TRY_CAST(c AS DOUBLE) BETWEEN 800 AND 999.99 THEN 'Injury'
WHEN TRY_CAST(c AS DOUBLE) BETWEEN 710 AND 739.99 THEN 'Musculoskeletal'
WHEN TRY_CAST(c AS DOUBLE) BETWEEN 580 AND 629.99 OR c LIKE '788%' THEN 'Genitourinary'
WHEN TRY_CAST(c AS DOUBLE) BETWEEN 140 AND 239.99 THEN 'Neoplasms'
ELSE 'Other' END""")
```

The clean table adds a numeric age (the midpoint of each 10-year bracket), readable lookup descriptions, the three diagnosis groups, total prior visits, HbA1c and glucose results with `Not Tested` as a level, an HbA1c group used in Section 5 (Not Tested, Normal, High with medication changed, High without change), counts of diabetes drugs used and changed, and the 0/1 target `readmit_30`. The post-clean check confirms 69,970 rows and 69,970 unique patients, a 30-day readmission rate of **8.97%**, and no NULLs left except in the raw diagnosis codes, whose groups already say `Missing`.

3 SQL business questions

Notebook 03. Thirteen questions a hospital quality team would ask, answered in DuckDB SQL on the clean cohort.

Each query sits in its own variable (`q01` , `q02` , and so on) and is passed to a small `run()` helper that executes it, saves the result to `outputs/sql_results/` and returns it. The queries use CTEs, window functions (`RANK` , `FIRST_VALUE` , `ROW_NUMBER`), `QUALIFY` , `FILTER` , `PIVOT` and `UNPIVOT` , `ROLLUP` with `GROUPING` , macros and variables. A reporting view adds the bands used across all questions: discharge group, age band, prior visit bins, length of stay band and an HbA1c tested flag. The view is also saved as a Parquet file for the later notebooks and for Power BI.

3.1 Rates with confidence intervals

Many groups here are small, and the outcome is rare, so a plain rate can mislead. Three macros return a rate and its 95% Wilson score interval. Unlike the normal approximation, the Wilson interval stays between 0 and 100% even for a group of 150 patients with 3 readmissions. The overall rate is stored in a variable so any group can be compared against it.

Code 3.1 Rate and Wilson interval macros.

03_sql_business_questions.ipynb, cell 5

```
-- Rate as a percentage
CREATE OR REPLACE MACRO rate(x) AS
  ROUND(100 * AVG(x), 2);

-- 95% Wilson score interval, lower bound (z = 1.96)
CREATE OR REPLACE MACRO ci_lo(x) AS
  ROUND(100 * (
    AVG(x) + 1.9208 / COUNT(*)
    - 1.96 * SQRT(AVG(x) * (1 - AVG(x)) / COUNT(*) + 0.9604 / COUNT(*) ^ 2)
  ) / (1 + 3.8416 / COUNT(*)), 2);

-- 95% Wilson score interval, upper bound (z = 1.96)
CREATE OR REPLACE MACRO ci_hi(x) AS
  ROUND(100 * (
    AVG(x) + 1.9208 / COUNT(*)
    + 1.96 * SQRT(AVG(x) * (1 - AVG(x)) / COUNT(*) + 0.9604 / COUNT(*) ^ 2)
  ) / (1 + 3.8416 / COUNT(*)), 2);

-- Overall 30-day readmission rate, stored for comparisons
SET VARIABLE overall = (SELECT AVG(readmit_30) FROM enc);
```

3.2 Headline numbers

69,970

patients in the cohort

8.97%

readmitted within 30 days

4.27

average length of stay (days)

18.4%

had HbA1c measured

51.0%

on insulin

45.0%

had a diabetes medication
change

15.7

medications per stay on
average

26.0%

had any hospital visit in the
previous year

3.3 Age, diagnosis and discharge

Readmission rises with age, from under 5% for patients below 20 to 10.8% for patients aged 80 to 89. The groups under 30 are small, and their intervals are wide.

Table 3.1 Q2. 30-day readmission by age bracket, with 95% Wilson intervals.

age	patients	readmit30_pct	ci_low	ci_high	diff_vs_overall_pp	alos_days
[0-10)	153	1.96	0.67	5.61	-7.01	2.57
[10-20)	534	4.87	3.34	7.04	-4.10	2.87
[20-30)	1,121	7.40	6.01	9.09	-1.57	3.46
[30-40)	2,692	6.98	6.08	8.01	-1.99	3.63
[40-50)	6,828	7.41	6.81	8.06	-1.56	3.85
[50-60)	12,349	7.12	6.68	7.58	-1.85	3.97
[60-70)	15,683	9.00	8.57	9.46	+0.03	4.24
[70-80)	17,748	10.24	9.80	10.69	+1.27	4.51
[80-90)	11,102	10.76	10.20	11.35	+1.79	4.78
[90-100)	1,760	9.55	8.26	11.01	+0.57	4.82

By primary diagnosis, injury admissions come back most often (10.8%) and respiratory admissions least often (7.3%). Circulatory problems are the largest group at 30.6% of patients. The HbA1c testing column already shows the gap studied in Section 5: 35.9% of patients admitted for diabetes were tested, against about 10% for musculoskeletal and cancer admissions.

Table 3.2 Q3. Readmission by primary diagnosis group, ranked by rate.

diag_1_group	patients	share_pct	readmit30_pct	ci_low	ci_high	alos_days	a1c_tested_pct	risk_rank
Injury	4,692	6.7	10.78	9.93	11.70	4.50	12.30	1
Circulatory	21,383	30.6	9.67	9.28	10.07	4.17	19.07	2
Diabetes	5,748	8.2	9.12	8.40	9.89	4.21	35.86	3
Neoplasms	2,538	3.6	9.06	8.01	10.24	5.15	9.73	4
Genitourinary	3,440	4.9	8.98	8.07	9.98	4.03	15.58	5
Other	12,122	17.3	8.97	8.47	9.49	4.61	18.44	6
Musculoskeletal	4,064	5.8	8.39	7.58	9.28	3.83	9.89	7
Digestive	6,487	9.3	8.02	7.38	8.70	4.28	12.98	8
Respiratory	9,486	13.6	7.28	6.78	7.82	4.04	19.65	9

Discharge destination separates patients more than any diagnosis does. Patients transferred to another care facility (a skilled nursing facility, rehab, another hospital) are readmitted at 15.2%, more than twice the 7.0% of patients sent home. These are usually sicker, older patients, which Section 5 adjusts for.

Table 3.3 Q4. Readmission by discharge destination (26 discharge codes grouped into 6).

discharge_group	patients	share_pct	readmit30_pct	ci_low	ci_high	alos_days
Transferred to facility	13,604	19.4	15.16	14.56	15.77	5.58
Other	28	0.0	10.71	3.71	27.20	4.86
Left against medical advice	409	0.6	9.54	7.05	12.77	2.76
Home with services	8,362	12.0	9.52	8.91	10.17	5.50
Unknown	3,252	4.6	9.19	8.25	10.24	4.83
Home	44,315	63.3	6.95	6.71	7.19	3.61

3.4 Prior hospital use

This is the strongest pattern in the data. The query stacks inpatient and emergency visit counts with `UNION ALL`, then divides each group's rate by the rate of patients with no prior visits of that type using `FIRST_VALUE` over a window.

Code 3.2 Q5. Readmission by prior inpatient and emergency visits, with relative risk. 03_sql_business_questions.ipynb, cell 15

```

WITH utilization AS (
  SELECT
    'Inpatient'      AS visit_type,
    prior_inpatient  AS visits,
    COUNT(*)         AS patients,
    AVG(readmit_30)  AS r
  FROM enc
  GROUP BY prior_inpatient

  UNION ALL

  SELECT
    'Emergency',
    prior_emergency,
    COUNT(*),
    AVG(readmit_30)
  FROM enc
  GROUP BY prior_emergency
)
SELECT
  visit_type,
  visits,
  patients,
  ROUND(100 * r, 2)                                AS readmit30_pct,
  ROUND(r / FIRST_VALUE(r) OVER (PARTITION BY visit_type ORDER BY visits), 2) AS relative_risk_vs_none
FROM utilization
ORDER BY visit_type, visits

```

Table 3.4 Q5. Prior visits in the year before the stay and 30-day readmission.

visit_type	visits	patients	readmit30_pct	relative_risk_vs_none
Emergency	0	64,870	8.73	1.00
Emergency	1	3,882	11.26	1.29
Emergency	2+	1,218	14.45	1.65
Inpatient	0	61,779	8.12	1.00
Inpatient	1	5,794	12.89	1.59
Inpatient	2	1,501	18.52	2.28
Inpatient	3+	896	26.45	3.26

Patients with three or more inpatient stays in the previous year are readmitted at 26.5%, 3.26 times the rate of patients with none. Emergency visits show the same direction, but weaker (1.65 times for two or more).

3.5 Length of stay

Table 3.5 Q6a. Readmission by length of stay band.

los_band	patients	readmit30_pct	ci_low	ci_high	avg_medications	avg_lab_procedures
1-2 days	22,646	7.14	6.82	7.49	11.6	35.4
3-4 days	21,910	8.48	8.11	8.85	14.8	41.6
5-7 days	15,649	10.54	10.07	11.03	18.3	48.4
8-14 days	9,765	11.80	11.17	12.45	22.9	54.3

Longer stays come with more readmissions (7.1% for one or two days, 11.8% for eight days or more), but also with twice the medications and more lab work. Length of stay is partly a marker of how sick the patient was.

3.6 HbA1c testing

The query below uses `FILTER` to compute the tested and untested rates side by side in one pass. The gap is the tested rate minus the untested rate, in percentage points.

Code 3.3 Q7. HbA1c testing rate and readmission for tested vs untested patients, by diagnosis. 03_sql_business_questions.ipynb, cell 20

```

SELECT
    diag_l_group,
    COUNT(*)
    rate(alc_tested)
    ROUND(100 * AVG(readmit_30) FILTER (WHERE alc_tested = 1), 2)
    ROUND(100 * AVG(readmit_30) FILTER (WHERE alc_tested = 0), 2)
    ROUND(readmit_tested - readmit_not_tested, 2)
FROM enc
WHERE diag_l_group <> 'Missing'
GROUP BY diag_l_group
ORDER BY patients DESC
AS patients,
AS alc_tested_pct,
AS readmit_tested,
AS readmit_not_tested,
AS gap_pp

```

Table 3.6 Q7. HbA1c testing and raw readmission gap by primary diagnosis.

diag_1_group	patients	a1c_tested_pct	readmit_tested	readmit_not_tested	gap_pp
Circulatory	21,383	19.07	10.20	9.54	+0.66
Other	12,122	18.44	8.41	9.09	-0.68
Respiratory	9,486	19.65	5.53	7.71	-2.18
Digestive	6,487	12.98	7.48	8.10	-0.62
Diabetes	5,748	35.86	7.23	10.17	-2.94
Injury	4,692	12.30	7.28	11.28	-4.00
Musculoskeletal	4,064	9.89	9.20	8.30	+0.90
Genitourinary	3,440	15.58	10.07	8.78	+1.29
Neoplasms	2,538	9.73	10.12	8.95	+1.17

The raw gap changes sign across diagnoses. Tested patients do better in injury (4.0 points lower), diabetes (2.9 points lower) and respiratory admissions, but worse in circulatory, genitourinary and cancer admissions. A pattern like this usually means the groups being compared differ in other ways. These are unadjusted numbers, and Section 5 deals with the confounding.

3.7 Medication

The 21 drug columns are unpivoted to long format so every drug can be summarized in one query. Only drugs with at least 500 users are kept.

Code 3.4 Q9. Diabetes drug use and readmission, after UNPIVOT.

03_sql_business_questions.ipynb, cell 24

```

cols = con.sql("FROM enc LIMIT 0").columns
drugs = ', '.join(cols[cols.index('metformin'):cols.index('metformin_pioglitazone') + 1])
q09 = f"""
WITH long AS (
    UNPIVOT (SELECT readmit_30, {drugs} FROM enc)
    ON {drugs}
    INTO NAME drug VALUE dose
)
SELECT
    drug,
    COUNT(*) FILTER (WHERE dose <> 'No')                AS patients_on_drug,
    ROUND(100 * AVG((dose <> 'No')::INT), 2)             AS usage_pct,
    ROUND(100 * AVG((dose IN ('Up', 'Down'))::INT) FILTER (WHERE dose <> 'No'), 1) AS dose_changed_pct,
    ROUND(100 * AVG(readmit_30) FILTER (WHERE dose <> 'No'), 2) AS readmit_users,
    ROUND(100 * AVG(readmit_30) FILTER (WHERE dose = 'No'), 2) AS readmit_non_users
FROM long
GROUP BY drug
HAVING patients_on_drug >= 500
ORDER BY patients_on_drug DESC
"""

```

Table 3.7 Q9. Diabetes drugs with 500 or more users.

drug	patients_on_drug	usage_pct	dose_changed_pct	readmit_users	readmit_non_users
insulin	35,715	51.04	39.5	9.63	8.28
metformin	14,902	21.30	8.5	8.30	9.15
glipizide	9,006	12.87	10.5	9.74	8.86
glyburide	7,775	11.11	13.3	9.38	8.92
pioglitazone	5,262	7.52	4.9	8.72	8.99
rosiglitazone	4,660	6.66	4.4	8.80	8.98
glimepiride	3,697	5.28	9.9	8.76	8.98
repaglinide	917	1.31	10.8	12.43	8.93

Half the cohort is on insulin. Insulin users are readmitted at 9.6% against 8.3% for non-users, and patients whose insulin dose was lowered have the highest rate (10.6%). Dose changes go with more medications and longer stays, so this likely reflects how unstable the patient was rather than the drug itself.

Table 3.8 Q10. Readmission by insulin dose change.

insulin	patients	readmit30_pct	ci_low	ci_high	avg_medications	alos_days
Down	7,321	10.55	9.86	11.27	18.8	4.82
Up	6,777	9.84	9.16	10.57	19.6	5.21
Steady	21,617	9.26	8.88	9.65	16.1	4.30
No	34,255	8.28	7.99	8.58	13.9	3.95

3.8 Specialty and equity check

Admitting specialty is unknown for about half the patients, so Q11 covers the known half. Among the 10 largest specialties, nephrology has the highest rate (11.0%) and reconstructive orthopedics the lowest (6.6%). Q12 uses ROLLUP to check readmission and HbA1c testing by race and gender. Readmission rates are close across groups (8.1% to 9.2% for the large groups), but testing is not: 26.3% of Hispanic patients were tested against 17.5% of Caucasian patients.

Table 3.9 Q12. Readmission and HbA1c testing by race (race subtotals and grand total from ROLLUP).

race	patients	readmit30_pct	a1c_tested_pct	alos_days
AfricanAmerican	12,625	8.66	20.48	4.34
Asian	488	8.40	21.93	3.68
Caucasian	52,292	9.18	17.45	4.28
Hispanic	1,500	8.13	26.27	3.95
Other	1,149	6.88	21.93	4.16
Unknown	1,916	7.36	19.89	4.24
All	69,970	8.97	18.36	4.27

3.9 Where to focus first

A care-transition program cannot follow up every patient. Q13 builds segments from age band, prior inpatient visits and primary diagnosis (only segments with 300 or more patients), then ranks them by **excess readmissions**: how many more readmissions the segment had than the overall rate would predict. This ranks by total impact rather than by rate alone, so a large segment with a moderate rate can rank above a small one with a high rate.

Code 3.5 Q13. Segments ranked by excess readmissions.

03_sql_business_questions.ipynb, cell 32

```
WITH segments AS (
  SELECT
    age_band,
    prior_inpatient,
    diag_1_group,
    COUNT(*)      AS patients,
    AVG(readmit_30) AS r
  FROM enc
  GROUP BY age_band, prior_inpatient, diag_1_group
  HAVING COUNT(*) >= 300
)
SELECT
  age_band,
  prior_inpatient,
  diag_1_group,
  patients,
  ROUND(100 * r, 2)      AS readmit30_pct,
  ROUND(r / getvariable('overall'), 2) AS lift,
  ROUND(patients * (r - getvariable('overall'))):INT AS excess_readmissions
FROM segments
ORDER BY excess_readmissions DESC
LIMIT 10
```

Table 3.10 Q13. Top 10 segments by excess readmissions.

age_band	prior_inpatient	diag_1_group	patients	readmit30_pct	lift	excess_readmissions
60-79	1	Circulatory	973	13.87	1.55	48
60-79	0	Injury	1,853	10.74	1.20	33
80+	0	Circulatory	3,546	9.84	1.10	31
80+	0	Injury	1,041	11.82	1.32	30
60-79	0	Circulatory	10,507	9.24	1.03	28
60-79	1	Other	620	12.90	1.44	24
80+	0	Other	1,887	10.02	1.12	20
80+	1	Circulatory	380	13.95	1.55	19
60-79	1	Respiratory	333	13.21	1.47	14
60-79	0	Genitourinary	1,412	9.63	1.07	9

What the SQL shows

Readmission is concentrated in patients with prior inpatient stays, patients transferred to other facilities, older patients and longer stays. The single largest excess comes from patients aged 60 to 79 with one prior stay and a circulatory diagnosis (973 patients, 13.9%, 48 more readmissions than expected). HbA1c is measured for fewer than one in five patients, and the raw effect of testing is inconsistent across diagnoses.

4 Exploratory data analysis

Notebook 04. Which features separate readmitted patients from the rest, and by how much.

With about 70,000 patients almost every difference is statistically significant, so p-values say very little here. Features are ranked by effect size instead: the rank-biserial correlation for numeric features and Cramér's V for categorical ones.

4.1 Target

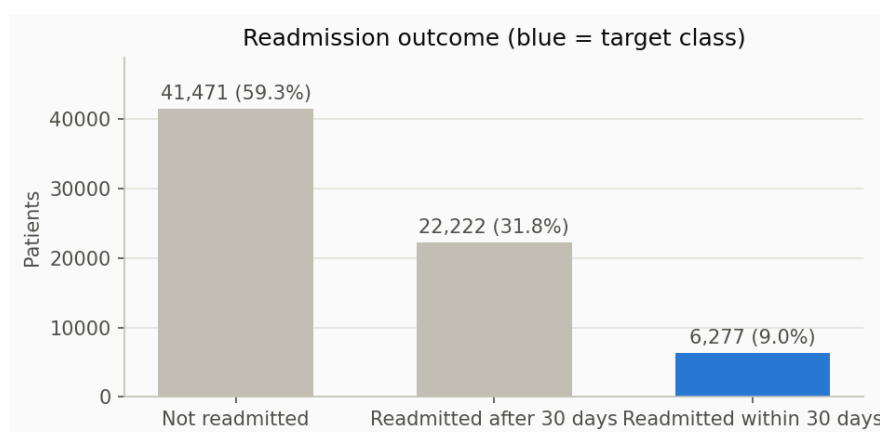


Figure 4.1 Readmission outcome in the cohort. The target class is readmission within 30 days.

Only 8.97% of patients are readmitted within 30 days. With a positive class this small, a model that predicts "no readmission" for everyone is 91% accurate and useless. Section 6 therefore uses PR-AUC, calibration and a cost-based threshold instead of accuracy.

4.2 Numeric features

The count features are right-skewed (most patients have zero prior visits), so the Mann-Whitney U test is used rather than a t-test. The rank-biserial r runs from -1 to 1. A positive value means readmitted patients tend to have higher values. As a rough guide, $|r|$ of 0.1 is small, 0.3 medium and 0.5 large.

Code 4.1 Mann-Whitney U test with rank-biserial effect size for each numeric feature.

04_edu.ipynb, cell 5

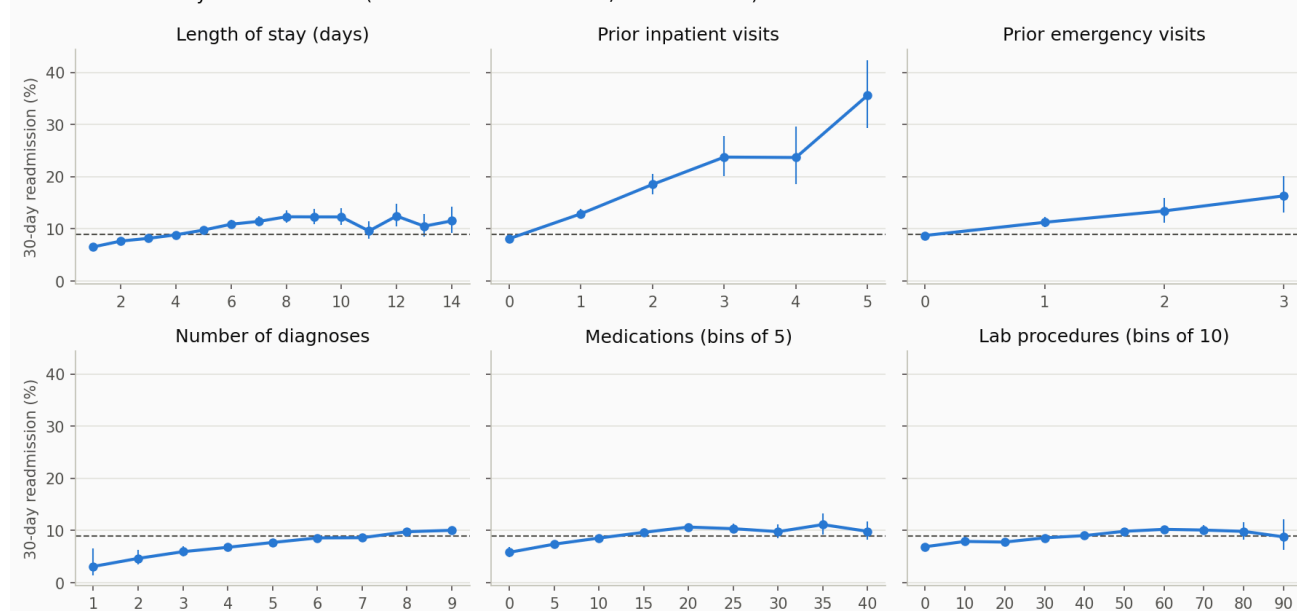
```
y = df.readmit_30 == 1
def mw(c):
    u, p = mannwhitneyu(df.loc[y, c], df.loc[~y, c])
    return pd.Series({'mean_readmitted': df.loc[y, c].mean(), 'mean_not': df.loc[~y, c].mean(),
                    'median_readmitted': df.loc[y, c].median(), 'median_not': df.loc[~y, c].median(), 'p_value': p,
                    'rank_biserial_r': 2*u/(y.sum()*(~y).sum()) - 1})
num = pd.DataFrame({c: mw(c) for c in NUM}).T.sort_values('rank_biserial_r', key=abs, ascending=False)
num.to_csv(TAB/'04_numeric_effects.csv'); num.round(4)
```

Every effect is small. Length of stay has the largest ($r = 0.12$), followed by age and prior inpatient visits. The mean of prior inpatient visits is more than twice as high for readmitted patients (0.37 vs 0.16), but both medians are zero, which is why the rank-based effect looks modest. The number of procedures shows no difference at all ($p = 0.35$).

Table 4.1 Numeric features ranked by effect size.

feature	mean_readmitted	mean_not	median_readmitted	median_not	p_value	rank_biserial_r
time_in_hospital	4.80	4.22	4.0	3.0	< 0.001	0.120
age_mid	67.83	65.20	75.0	65.0	< 0.001	0.096
number_inpatient	0.37	0.16	0.0	0.0	< 0.001	0.096
num_medications	16.63	15.57	15.0	14.0	< 0.001	0.086
number_diagnoses	7.51	7.20	8.0	8.0	< 0.001	0.083
num_lab_procedures	44.92	42.68	46.0	44.0	< 0.001	0.066
n_meds_changed	0.30	0.26	0.0	0.0	< 0.001	0.035
n_meds_used	1.22	1.19	1.0	1.0	< 0.001	0.028
number_emergency	0.15	0.10	0.0	0.0	< 0.001	0.028
number_outpatient	0.31	0.28	0.0	0.0	< 0.001	0.019
num_procedures	1.42	1.43	1.0	1.0	0.351	0.007

Readmission rate by numeric feature (dashed line = overall rate, bars = 95% CI)

**Figure 4.2** Readmission rate across each numeric feature. Dashed line is the overall rate, bars are 95% Wilson intervals. The last point on each panel includes all higher values.

4.3 Categorical features

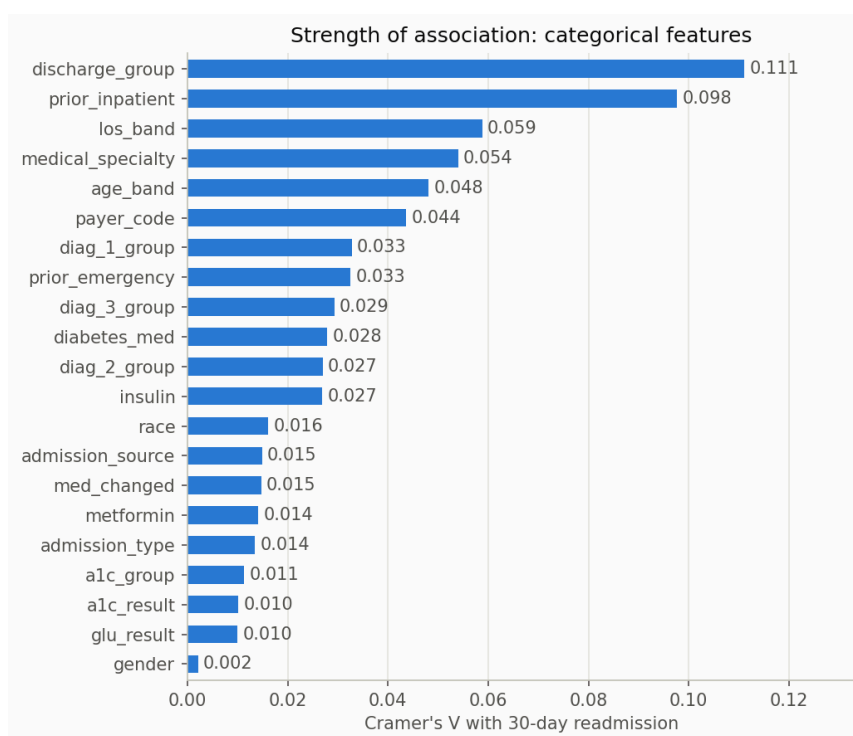


Figure 4.3 Cramér's V between each categorical feature and 30-day readmission.

Discharge group ($V = 0.111$) and prior inpatient visits ($V = 0.098$) lead by a clear margin. The HbA1c variables are near the bottom (V of about 0.01). `medical_specialty` and `payer_code` have many sparse levels, which inflates V a little, so their position should be read with that in mind.

30-day readmission by patient segment (groups with 100+ patients, dashed line = overall)

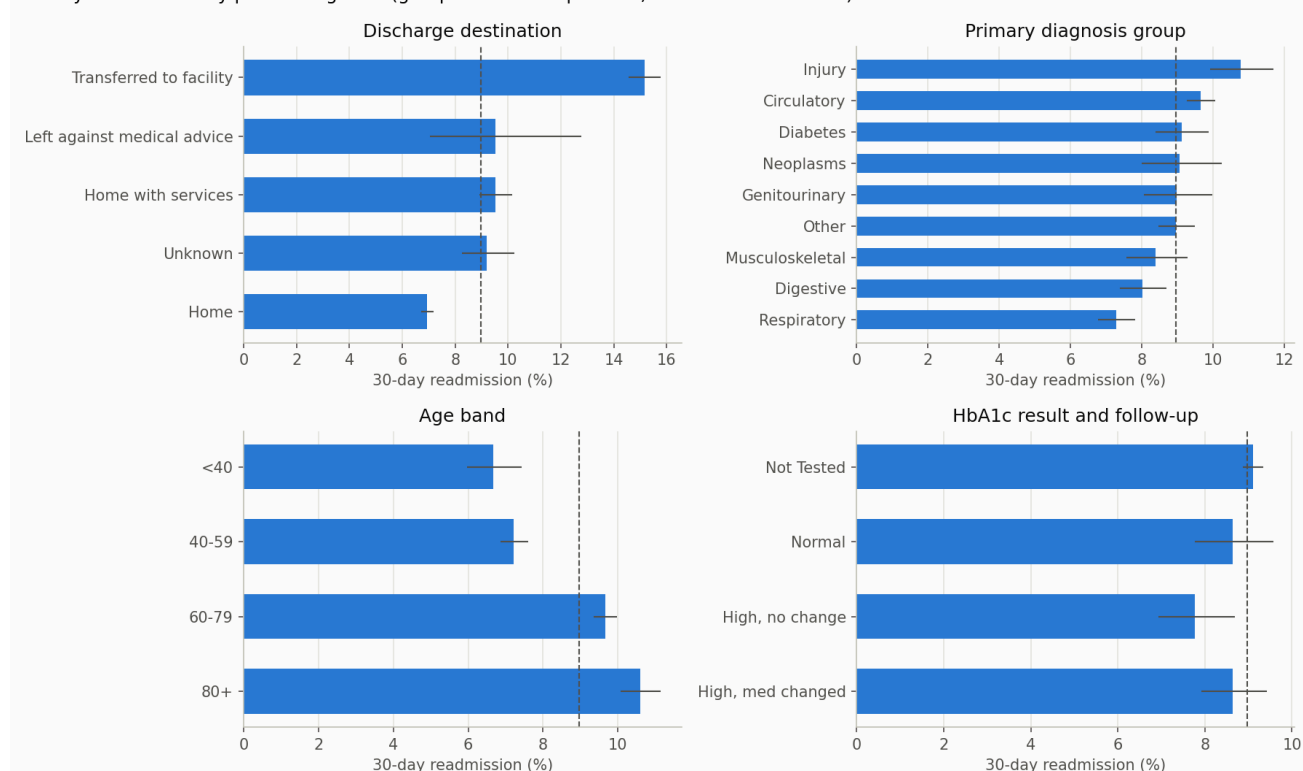


Figure 4.4 Readmission by patient segment, groups with 100 or more patients. Dashed line is the overall rate.

4.4 The HbA1c testing gap

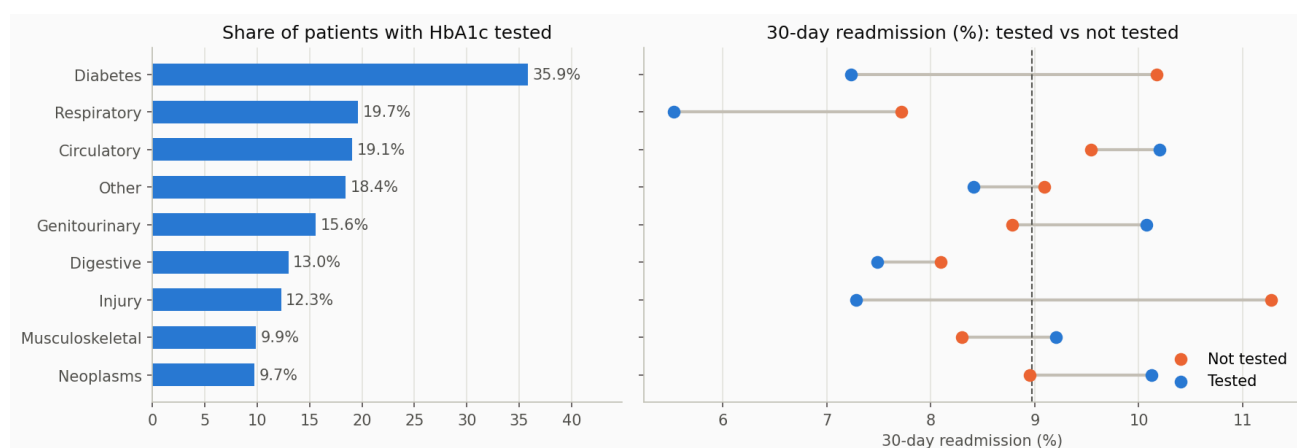


Figure 4.5 Left: share of patients with HbA1c measured, by primary diagnosis. Right: raw readmission for tested (blue) and untested (orange) patients.

Testing is low everywhere and very uneven. The dumbbell chart shows tested patients doing better in some diagnoses and worse in others. If testing itself helped, the direction should be more consistent. More likely, the patients who get tested differ from those who do not, and that difference changes from one diagnosis to the next.

4.5 Risk compounds with age

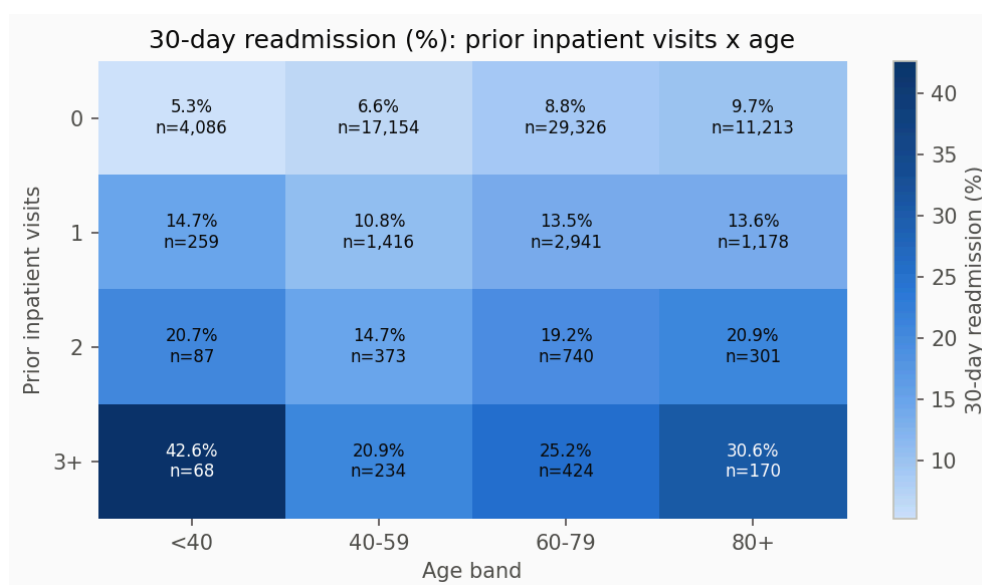


Figure 4.6 30-day readmission by prior inpatient visits and age band. Each cell shows the rate and the number of patients.

Without prior stays, readmission climbs slowly with age, from 5.3% to 9.7%. With three or more prior stays it sits between 20.9% and 30.6% for patients over 40, and reaches 42.6% for the 68 patients under 40. That last cell is small, but the direction matches every other row: prior inpatient use is the main multiplier of risk at any age.

4.6 Correlation between features

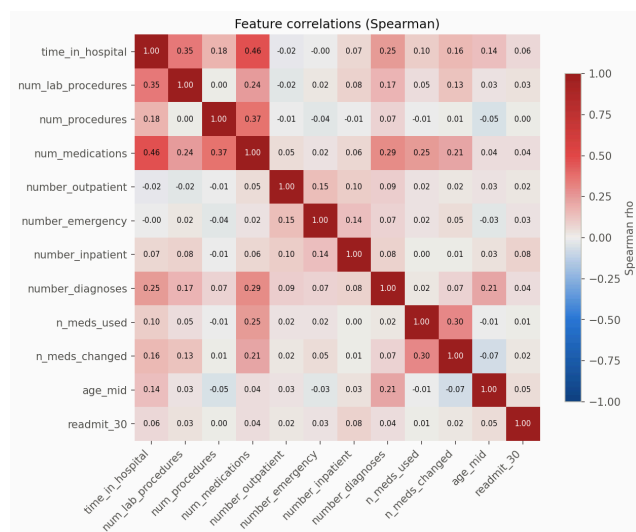


Figure 4.7 Spearman correlation between numeric features and the target.

Only one pair reaches $|\rho|$ of 0.4: length of stay and number of medications ($\rho = 0.46$). Longer stays mean more drugs, which makes sense. Nothing is close to the level that would break a regression, and the variance inflation check in Section 5 confirms it.

The correlations with `readmit_30` in the last row are all below 0.1, which matches the small effect sizes above. No single feature predicts readmission well on its own.

Takeaways for modeling

- The target is imbalanced at about 9%. Use PR-AUC, calibration and class-aware thresholds.
- Prior inpatient visits and discharge destination are the strongest signals. Age and length of stay add steady but smaller increases.
- HbA1c testing is rare and its raw link to readmission changes direction across diagnoses, a sign of confounding.
- Sparse categories in specialty, payer and admission source need grouping before modeling.

5 Statistical inference: HbA1c testing

Notebook 05. Is HbA1c testing linked to lower 30-day readmission once patient differences are accounted for, and does it depend on the reason for admission?

Section 4 showed that the raw tested vs untested gap flips direction across diagnoses. Three approaches are used to separate the effect of testing from the effect of who gets tested, and then checked against each other: an adjusted logistic regression, an interaction model with a likelihood-ratio test, and propensity score matching.

5.1 Modeling frame

Sparse categories are grouped first so every level has enough patients for a stable estimate. Admission source becomes Emergency, Referral, Transfer and Other/Unknown. Specialty keeps the four largest plus Surgery, Unknown and Other. Race becomes Caucasian, African American and Other/Unknown. Reference levels are age under 40, female, Caucasian, admitted through emergency, discharged home, no prior inpatient or emergency visits, circulatory diagnosis and internal medicine.

One choice matters more than the rest. Medication change and diabetes medication are **not** adjusted for. They happen after the HbA1c result and can be caused by it, so they are mediators, not confounders. Adjusting for them would remove part of the effect being measured.

Code 5.1 Covariate list and the unadjusted and adjusted logistic models. 05_statistical_inference.ipynb, cell 3 and cell 5

```
COV = ["C(age_band, Treatment('<40'))", 'C(gender)', "C(race_grp, Treatment('Caucasian'))", "C(admission_src, Treatment('Emergency'))", "C(discharge, Treatment('Home'))",
      'time_in_hospital', "C(prior_inpatient, Treatment('0'))", "C(prior_emergency, Treatment('0'))",
      'any_outpatient', 'number_diagnoses', 'num_medications',
      "C(diag, Treatment('Circulatory'))", "C(specialty, Treatment('InternalMedicine'))"]
crude = smf.logit('readmit_30 ~ alc_tested', m).fit(dis=0)
adj = smf.logit('readmit_30 ~ alc_tested + ' + ' + '.join(COV), m).fit(dis=0)
```

5.2 Unadjusted vs adjusted effect

Table 5.1 Odds ratio of 30-day readmission for HbA1c tested vs not tested.

method	OR	ci_low	ci_high	p_value
Unadjusted logistic	0.915	0.854	0.980	0.011
Adjusted logistic	0.955	0.889	1.026	0.206

Without adjustment, tested patients have 8.5% lower odds of readmission (OR 0.915, $p = 0.011$). After adjusting for age, sex, race, admission source, discharge, length of stay, prior visits, number of diagnoses, medications, diagnosis and specialty, the odds ratio moves to 0.955 and the interval includes 1 ($p = 0.21$). About half of the raw difference was patient mix.

5.3 The full adjusted model

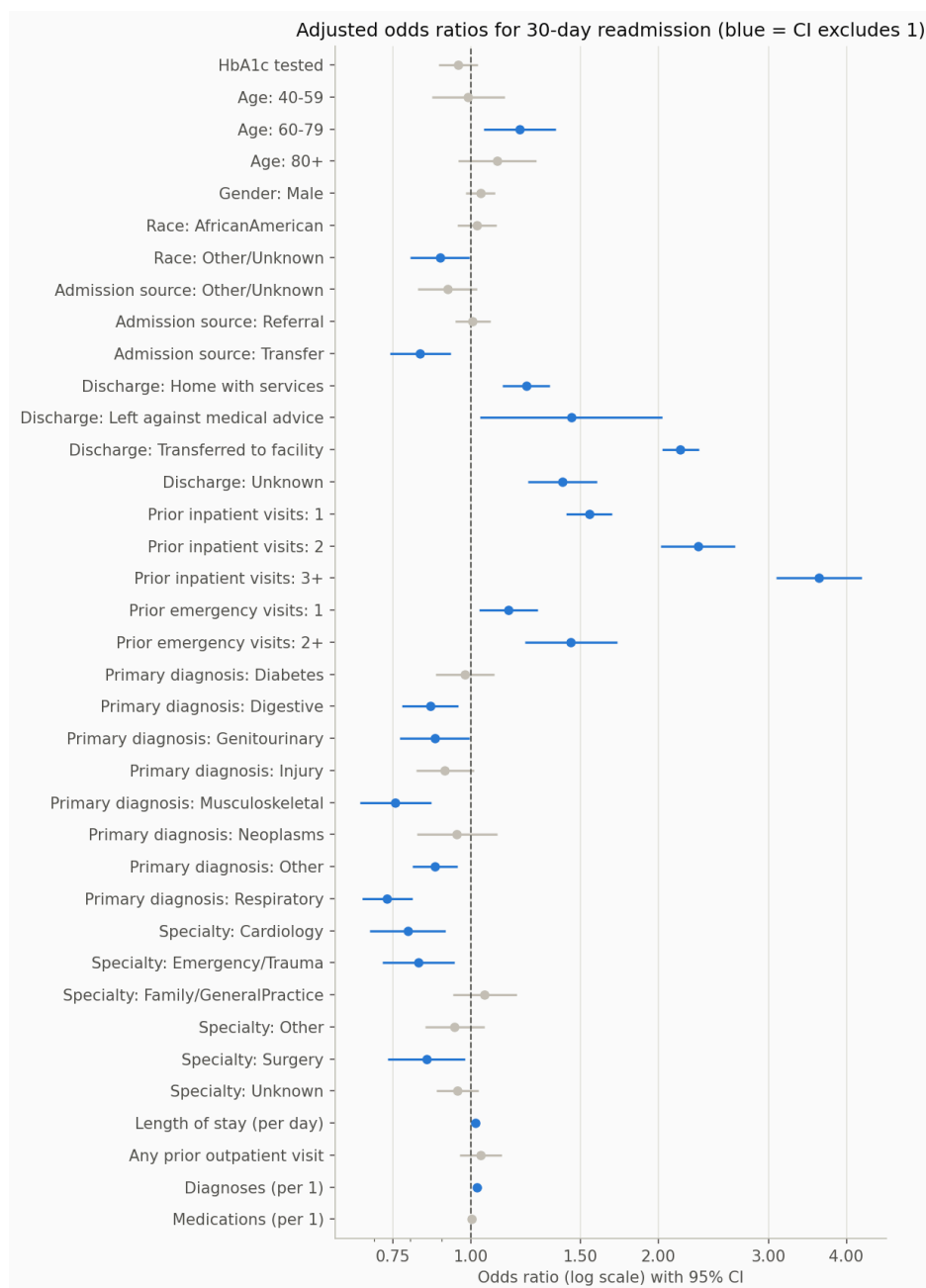


Figure 5.1 Adjusted odds ratios for 30-day readmission, log scale. Blue means the 95% interval excludes 1.

Table 5.2 Covariates with an adjusted odds ratio significantly above 1.

label	OR	ci_low	ci_high	p_value
Prior inpatient visits: 3+	3.61	3.09	4.23	< 0.001
Prior inpatient visits: 2	2.31	2.02	2.65	< 0.001
Discharge: Transferred to facility	2.17	2.03	2.32	< 0.001
Prior inpatient visits: 1	1.55	1.42	1.68	< 0.001
Discharge: Left against medical advice	1.45	1.04	2.03	0.030
Prior emergency visits: 2+	1.45	1.22	1.72	< 0.001
Discharge: Unknown	1.40	1.24	1.60	< 0.001
Discharge: Home with services	1.23	1.13	1.34	< 0.001
Age: 60-79	1.20	1.05	1.37	0.007
Prior emergency visits: 1	1.15	1.03	1.28	0.011
Diagnoses (per 1)	1.02	1.01	1.04	0.002
Length of stay (per day)	1.02	1.01	1.03	< 0.001

The regression agrees with the SQL. Three or more prior inpatient stays multiply the odds of readmission by 3.6, and a transfer to another facility by 2.2, after everything else is held fixed. Respiratory, musculoskeletal and digestive admissions carry lower odds than circulatory ones. Each extra day in hospital adds about 1.8% to the odds.

Checks. The largest variance inflation factor is 5.16, for the age 60 to 79 dummy, which is expected because the age dummies share one reference group. Every other VIF is below 4.2. The model's C-statistic is 0.643 and McFadden's pseudo R^2 is 0.035. The Hosmer-Lemeshow test rejects ($\chi^2 = 37.1$, $df = 8$, $p < 0.001$), but with 70,000 patients even small gaps become significant. Observed and expected rates by decile stay close, from 3.8% vs 4.7% in the lowest decile to 18.1% vs 19.1% in the highest.

5.4 Does the effect depend on the diagnosis?

An HbA1c by primary diagnosis interaction is added, and the model is compared with the main-effects model by a likelihood-ratio test. The odds ratio of testing within each diagnosis is then estimated with a linear contrast of the coefficients.

Code 5.2 Interaction model, likelihood-ratio test and per-diagnosis contrasts.

05_statistical_inference.ipynb, cell 14

```

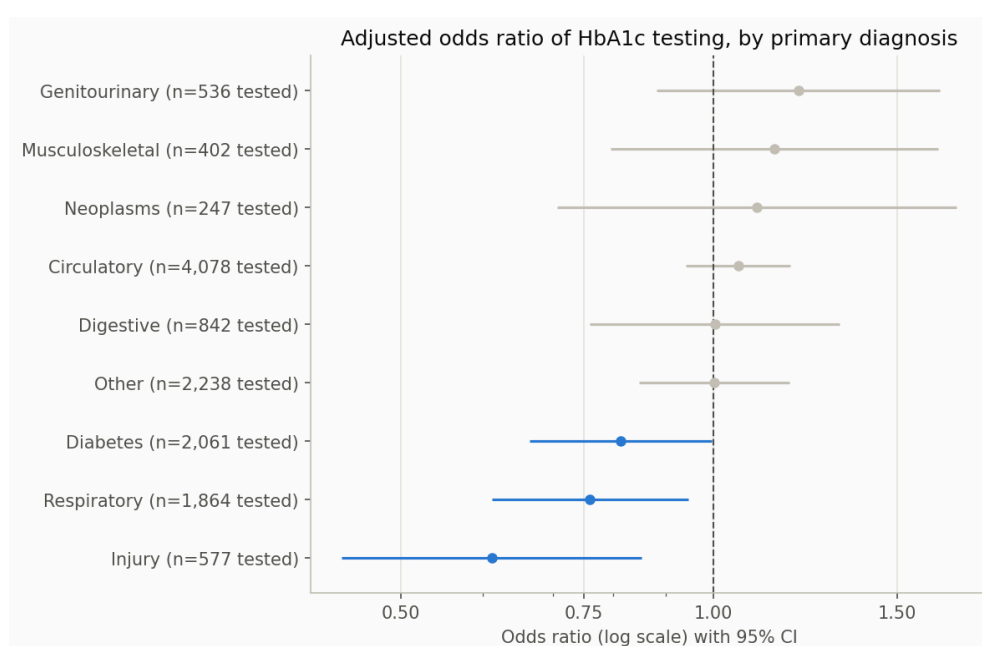
cov_wo_diag = [c for c in COV if not c.startswith('C(diag)')]
inter = smf.logit("readmit_30 ~ alc_tested * C(diag, Treatment('Circulatory')) + " + ' + '.join(cov_wo_diag),
m).fit(disp=0)
lr, ddf = 2*(inter.llf - adj.llf), inter.df_model - adj.df_model
names, rows = inter.params.index, []
for grp in sorted(m.diag.unique()):
    L = np.zeros(len(names)); L[names.get_loc('alc_tested')] = 1
    L[[i for i, n in enumerate(names) if ':' in n and 'alc_tested' in n and n.endswith(f'[T.{grp}]')]] = 1
    r = inter.t_test(L); lo, hi = r.conf_int()[0]
    rows.append({'label': grp, 'patients': int(m.diag.eq(grp).sum()), 'tested': int(m.loc[m.diag.eq(grp),
'alc_tested'].sum()), 'OR': np.exp(r.effect[0]), 'ci_low': np.exp(lo), 'ci_high': np.exp(hi), 'p_value':
float(np.squeeze(r.pvalue))})

```

The interaction is significant (LR $\chi^2 = 21.17$, $df = 8$, $p = 0.0067$). The effect of testing is not the same across diagnoses.

Table 5.3 Adjusted odds ratio of HbA1c testing within each primary diagnosis.

label	patients	tested	OR	ci_low	ci_high	p_value
Circulatory	21,383	4,078	1.055	0.940	1.185	0.360
Other	12,132	2,238	1.001	0.848	1.183	0.987
Respiratory	9,486	1,864	0.761	0.612	0.946	0.014
Digestive	6,487	842	1.003	0.761	1.322	0.984
Diabetes	5,748	2,061	0.814	0.666	0.995	0.045
Injury	4,692	577	0.612	0.439	0.852	0.004
Musculoskeletal	4,064	402	1.144	0.796	1.644	0.467
Genitourinary	3,440	536	1.206	0.882	1.650	0.241
Neoplasms	2,538	247	1.101	0.708	1.713	0.670

**Figure 5.2** Adjusted odds ratio of HbA1c testing by primary diagnosis. Blue means the interval excludes 1.

Testing is linked to lower odds of readmission in three groups: injury (OR 0.61, 95% CI 0.44 to 0.85), respiratory (0.76, 0.61 to 0.95) and diabetes (0.81, 0.67 to 1.00). For circulatory admissions, the largest group, there is no link (1.06, 0.94 to 1.19). This matches the direction reported by Strack et al. (2014), who also found that the relationship depended on the primary diagnosis.

5.5 Propensity score matching

The regression results depend on the model being correctly specified. Matching is a design-based check that does not. Each patient's probability of being tested is modelled from the same covariates. Every tested patient is then matched to the untested patient with the closest logit propensity score, within a caliper of 0.2 standard deviations, and balance is checked with standardized mean differences (SMD).

Code 5.3 Propensity model and 1:1 nearest-neighbor matching on the logit propensity score.

05_statistical_inference.ipynb, cell 17

```
ps_mod = smf.logit('alc_tested ~ ' + ' + '.join(COV), m).fit(dis=0)
m['ps'] = ps_mod.predict(m); m['lps'] = np.log(m.ps/(1 - m.ps))
caliper = 0.2*m.lps.std()
T, U = m[m.alc_tested == 1], m[m.alc_tested == 0]
dist, idx = NearestNeighbors(n_neighbors=1).fit(U[['lps']]).kneighbors(T[['lps']])
ok = dist[:, 0] <= caliper
pairs = pd.DataFrame({'t': T.index[ok], 'c': U.index[idx[ok, 0]]})
```

The propensity model has an AUC of 0.673, so testing is only partly predictable from patient characteristics. All 12,845 tested patients found a match within the caliper (0.129), using 10,216 unique untested patients.

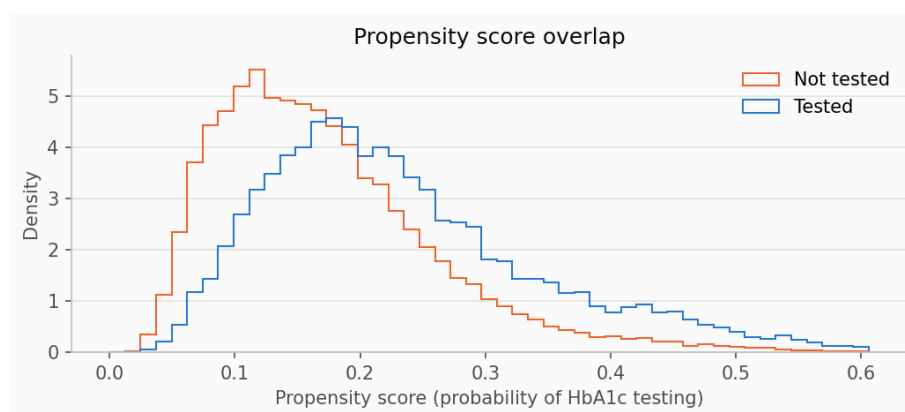


Figure 5.3 Propensity score distributions for tested and untested patients.

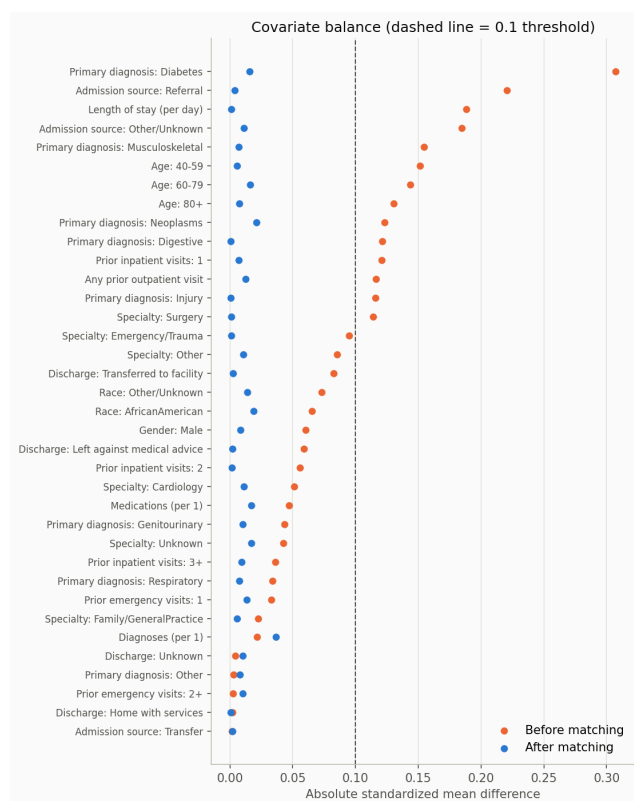


Figure 5.4 Covariate balance before and after matching. The dashed line is the usual 0.1 threshold.

The two groups overlap well across the whole score range, so every tested patient has comparable untested patients.

Before matching, the largest imbalance was the diabetes diagnosis (SMD 0.307): patients admitted for diabetes are far more likely to be tested. Admission through referral (0.221) and length of stay (0.188) were next.

After matching, the largest $|SMD|$ is 0.037 and no covariate is above 0.1. The matched groups are comparable on everything the model knows about.

Code 5.4 Readmission in the matched sample with 1,000 bootstrap resamples.

05_statistical_inference.ipynb, cell 20

```
yt, yc = m.loc[pairs.t, 'readmit_30'].to_numpy(), m.loc[pairs.c, 'readmit_30'].to_numpy()
odds_ratio = lambda a, b: (a.mean()/(1 - a.mean()))/(b.mean()/(1 - b.mean()))
rng = np.random.default_rng(42); boot = np.array([(yt[i].mean() - yc[i].mean()), odds_ratio(yt[i], yc[i])] for i
in (rng.integers(0, len(yt), len(yt)) for _ in range(1000))])
```

In the matched sample, readmission is 8.39% for tested patients and 8.49% for their matches, a difference of -0.09 percentage points (95% bootstrap CI -0.77 to 0.59).

Table 5.4 Effect of HbA1c testing across the three methods.

method	OR	ci_low	ci_high	p_value
Unadjusted logistic	0.915	0.854	0.980	0.011
Adjusted logistic	0.955	0.889	1.026	0.206
Propensity score matched	0.988	0.904	1.079	

What the statistics show

Across all patients, HbA1c testing is not linked to 30-day readmission once patient mix is adjusted for. The adjusted and matched estimates agree (OR 0.96 and 0.99), and both intervals include 1. The crude benefit came from who gets tested. But the effect differs by diagnosis, and for injury, respiratory and diabetes admissions testing goes with 19% to 39% lower odds of readmission.

This is observational data, so it is an association, not proof that testing prevents readmission. Testing may also stand for a more careful team or a more thorough work-up. It is still a reasonable place to look: a cheap test, rarely done, linked to fewer readmissions in three specific groups.

6 Readmission risk model

Notebook 06. Scoring every patient's risk of readmission within 30 days so a care team can target follow-up.

Section 5 estimated the effect of one factor. This section is a prediction task, so every feature known at discharge can be used, including medication changes. IDs, raw ICD-9 codes (the grouped versions are used instead) and the outcome columns are left out.

6.1 Setup

There are 14 categorical and 13 numeric features. The data is split 80/20 with stratification, so both parts have the same 8.97% readmission rate (55,976 training and 13,994 test patients). The test set is used once, at the very end. Categorical levels with fewer than 100 patients are merged into an "infrequent" level inside the one-hot encoder, so rare categories cannot cause overfitting and unseen ones cannot break the pipeline.

Code 6.1 Features, stratified split and preprocessing.

06_readmission_model.ipynb, cell 3

```
CAT = ['gender', 'race', 'admission_type', 'admission_source', 'discharge_disposition', 'payer_code',
       'medical_specialty',
       'diag_1_group', 'diag_2_group', 'diag_3_group', 'alc_result', 'glu_result', 'insulin', 'metformin']
NUM = ['age_mid', 'time_in_hospital', 'num_lab_procedures', 'num_procedures', 'num_medications',
       'number_outpatient', 'number_emergency',
       'number_inpatient', 'number_diagnoses', 'n_meds_used', 'n_meds_changed', 'med_changed', 'diabetes_med']
X, y = df[CAT + NUM], df.readmit_30
X_tr, X_te, y_tr, y_te = train_test_split(X, y, test_size=0.2, stratify=y, random_state=SEED)
pre = ColumnTransformer([('cat', OneHotEncoder(handle_unknown='infrequent_if_exist', min_frequency=100,
sparse_output=False), CAT), ('num', StandardScaler(), NUM)])
cv3, cv5 = StratifiedKFold(3, shuffle=True, random_state=SEED), StratifiedKFold(5, shuffle=True,
random_state=SEED)
```

6.2 Models and tuning

Two models are compared: logistic regression as an interpretable baseline, and XGBoost to capture non-linear effects and interactions. Both are tuned with randomized search and 3-fold cross-validation, scored by PR-AUC (average precision), which suits a 9% positive class much better than accuracy or ROC-AUC alone.

Code 6.2 XGBoost pipeline and randomized hyperparameter search.

06_readmission_model.ipynb, cell 7

```
xgb_params = {'clf__n_estimators': [200, 400, 600], 'clf__learning_rate': [0.02, 0.05, 0.1], 'clf__max_depth':
[3, 4, 5], 'clf__min_child_weight': [1, 5, 10],
              'clf__subsample': [0.7, 0.85, 1.0], 'clf__colsample_bytree': [0.6, 0.8, 1.0], 'clf__reg_lambda':
[1, 5, 10]}
xgb_search = RandomizedSearchCV(Pipeline([('pre', pre), ('clf', xgb.XGBClassifier(tree_method='hist',
eval_metric='logloss', random_state=SEED, n_jobs=-1))]), xgb_params,
                               n_iter=15, scoring='average_precision', cv=cv3, random_state=SEED,
n_jobs=1).fit(X_tr, y_tr)
print(f'XGBoost: CV PR-AUC = {xgb_search.best_score_:.4f}')
```

The best XGBoost setting uses 600 trees of depth 5, a learning rate of 0.02 and 70% row subsampling. Its cross-validated PR-AUC is 0.186, against 0.165 for logistic regression (best C = 0.248).

The final model is picked on **out-of-fold** predictions: every training patient gets a score from a model that never saw them, through 5-fold cross-validation. The same predictions are later used to choose the threshold. This keeps the test set completely clean.

Code 6.3 Model selection on out-of-fold predictions.

06_readmission_model.ipynb, cell 9

```
models = {'Logistic regression': lr_search.best_estimator_, 'XGBoost': xgb_search.best_estimator_}
oof = {k: cross_val_predict(v, X_tr, y_tr, cv=cv5, method='predict_proba')[:, 1] for k, v in models.items()}
best = max(oof, key=lambda k: average_precision_score(y_tr, oof[k]))
```

Table 6.1 Cross-validated and out-of-fold scores on the training set. XGBoost is selected.

model	cv_pr_auc (tuning)	oof_roc_auc	oof_pr_auc
Logistic regression	0.1646	0.6444	0.1635
XGBoost	0.1860	0.6580	0.1868

6.3 Test set performance

Table 6.2 Test set results (13,994 patients). Intervals are from 500 bootstrap resamples. The PR-AUC baseline is the readmission rate, which is what a random model would score.

model	roc_auc	roc_auc_ci	pr_auc	pr_auc_ci	pr_auc_baseline	brier	log_loss
Logistic regression	0.651	0.635 to 0.669	0.172	0.155 to 0.189	0.090	0.0790	0.2884
XGBoost	0.659	0.643 to 0.674	0.183	0.167 to 0.202	0.090	0.0786	0.2866

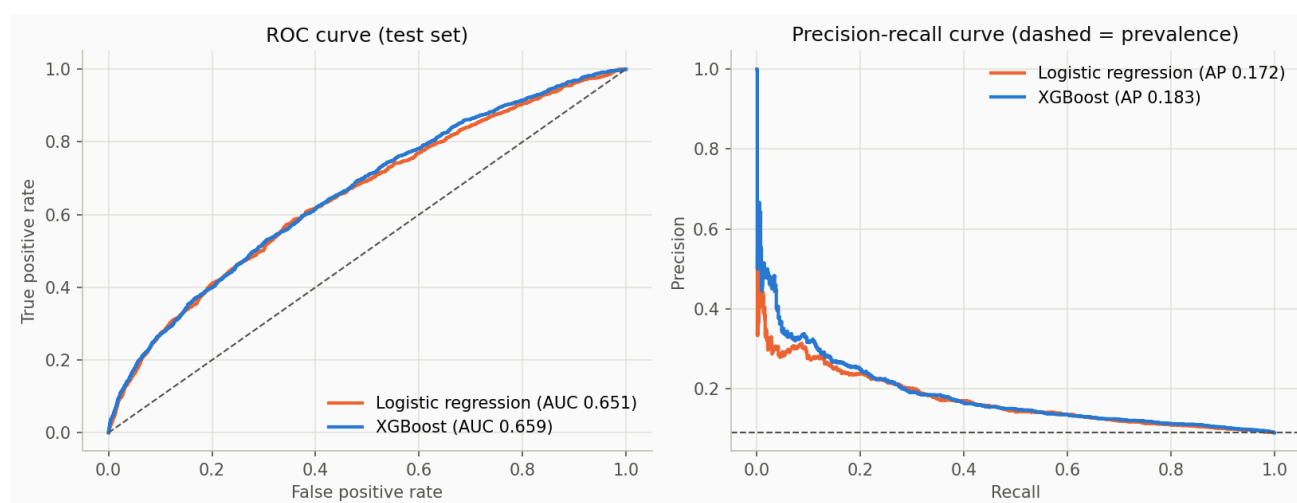


Figure 6.1 ROC and precision-recall curves on the test set.

XGBoost reaches a test ROC-AUC of **0.659** (0.643 to 0.674) and a PR-AUC of **0.183**, about twice the 0.090 a random model would get. It beats logistic regression on every metric, but not by much, and the intervals overlap. The test scores are close to the out-of-fold ones, so the model is not overfitting.

This level of discrimination is modest, and it is worth saying so plainly. The data has no information on things that are known to drive readmission, such as social support at home, follow-up appointments after discharge, or whether the patient takes the medication. No model can learn those from this data.

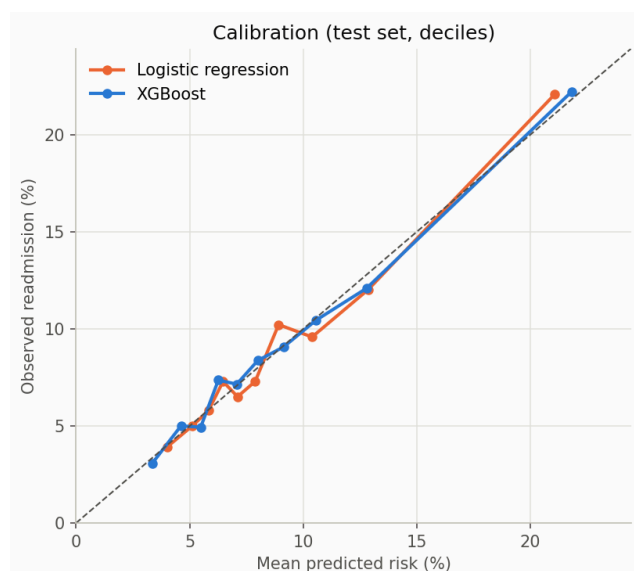


Figure 6.2 Calibration on the test set, by decile of predicted risk.

Discrimination is only half of it. If the model says 20%, about 20% of those patients should be readmitted, otherwise the risk tiers and the cost threshold below cannot be trusted.

Both models follow the diagonal closely across all deciles. XGBoost has the lower Brier score (0.0786 vs 0.0790). The scores can be read as real probabilities, which is what makes the rest of this section work.

6.4 Choosing a decision threshold

A missed readmission costs much more than an unnecessary follow-up call. The ratio is set to 10 to 1 as an **assumption** that a hospital would replace with its own numbers. The threshold with the lowest total expected cost is chosen on out-of-fold training predictions and only then applied to the test set.

Code 6.4 Cost-based threshold search.

06_readmission_model.ipynb, cell 16

```
COST_FP, COST_FN = 1, 10
p_oof, th = oof[best], np.round(np.arange(0.02, 0.50, 0.005), 3)
cost = np.array([COST_FP*((p_oof >= t) & (y_tr == 0)).sum() + COST_FN*((p_oof < t) & (y_tr == 1)).sum() for t
in th])
t_star = th[cost.argmin()]
```

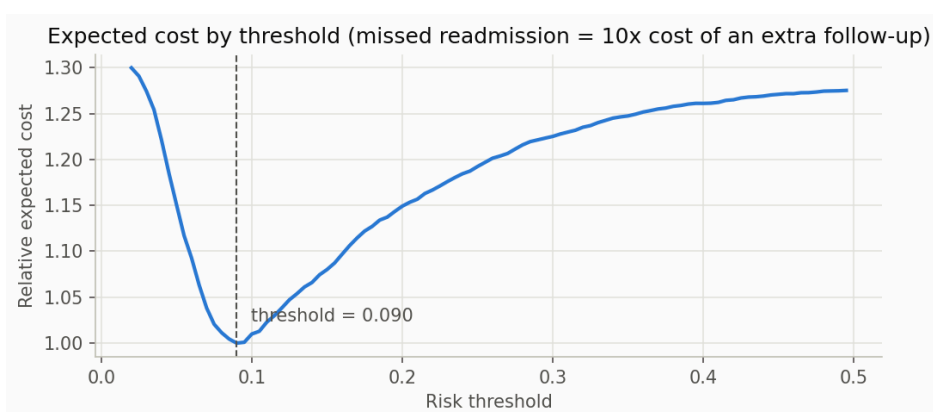


Figure 6.3 Relative expected cost by threshold, with a missed readmission costing 10 times an extra follow-up.

The best threshold is 0.090. For a well-calibrated model the theory gives $1 / (1 + 10) = 0.091$, so the search and the theory agree, which is another sign the scores are well calibrated.

Table 6.3 Test set confusion matrix at threshold 0.090.

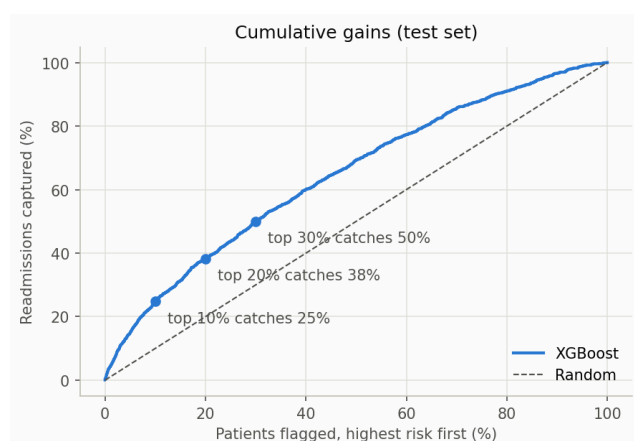
Actual	Flagged high risk	Not flagged
Readmitted	699	556
Not readmitted	4,331	8,408

Table 6.4 Test set metrics at threshold 0.090.

metric	value
Threshold	0.090
Patients flagged	35.9%
Sensitivity (recall)	55.7%
Specificity	66.0%
Precision (PPV)	13.9%
Negative predictive value	93.8%

At this threshold the model flags 36% of patients and catches 56% of readmissions (699 of 1,255). One in seven flagged patients is actually readmitted, which is 1.5 times the base rate. Among patients not flagged, 93.8% are not readmitted.

6.5 Ranking patients

**Figure 6.4** Cumulative gains on the test set.

A care team with limited capacity would work down the list from the highest-risk patient. The gains curve shows how many readmissions that reaches.

The top 10% of patients by predicted risk hold about a quarter of the readmissions, and the top 20% hold 38%, almost twice what a random list would give.

Table 6.5 Risk deciles for all 69,970 patients, using out-of-sample scores. Decile 10 is the highest risk.

risk_decile	patients	mean_predicted	observed_rate	readmissions	cum_capture
10.0	6,997	22.0%	22.2%	1,552	24.7%
9.0	6,997	12.9%	12.2%	855	38.3%
8.0	6,997	10.6%	10.7%	751	50.3%
7.0	6,997	9.1%	8.9%	622	60.2%
6.0	6,997	8.0%	8.2%	573	69.3%
5.0	6,997	7.0%	6.9%	486	77.1%
4.0	6,997	6.2%	6.3%	444	84.2%
3.0	6,997	5.4%	5.9%	411	90.7%
2.0	6,997	4.6%	4.9%	340	96.1%
1.0	6,997	3.3%	3.5%	243	100.0%

6.6 What drives the predictions

SHAP splits each prediction into the contribution of every feature. XGBoost computes exact tree SHAP values directly through `pred_contribs=True`. The one-hot columns are then summed back to their original feature with a mapping matrix, so importance is reported per variable (for example "Discharge destination"), not per category.

Code 6.5 SHAP contributions summed back to the original features.

06_readmission_model.ipynb, cell 21

```
final = models[best]
Xt_all = pd.DataFrame(final.named_steps['pre'].transform(X),
                      columns=final.named_steps['pre'].get_feature_names_out(), index=X.index)
ohe_names = final.named_steps['pre'].named_transformers_['cat'].get_feature_names_out()
orig = [next(c for c in CAT if n.startswith(c + '_')) for n in ohe_names] + NUM
if best == 'XGBoost': contrib = final.named_steps['clf'].get_booster().predict(xgb.DMatrix(Xt_all.to_numpy()),
                                     pred_contribs=True)[: , :-1]
else: contrib = (Xt_all.to_numpy() -
                 Xt_all.loc[X_tr.index].mean().to_numpy())*final.named_steps['clf'].coef_[0]
A = pd.get_dummies(pd.Series(orig)).reindex(columns=CAT + NUM).to_numpy(float)
S = contrib @ A
imp = pd.Series(np.abs(S).mean(0), index=[LABELS[c] for c in CAT + NUM]).sort_values(ascending=False)
```

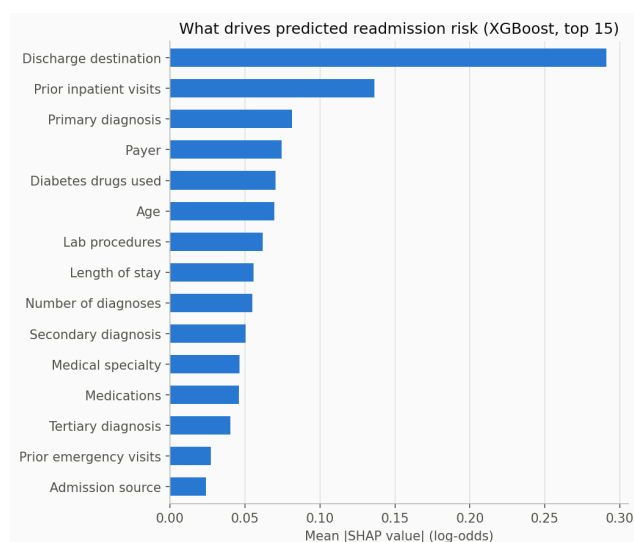


Figure 6.5 Mean absolute SHAP value per feature, top 15.

Discharge destination is the strongest driver by a wide margin (mean $|\text{SHAP}|$ 0.29), twice the next one, prior inpatient visits (0.14). Primary diagnosis, payer, number of diabetes drugs and age follow.

This is the same ranking the SQL, the EDA and the regression gave. Four different methods pointing at the same two features is a good sign that the pattern is real and not an artifact of one method.

HbA1c result ranks 21st of 27. It helps explain readmission in specific diagnoses (Section 5) but adds little to prediction across all patients.

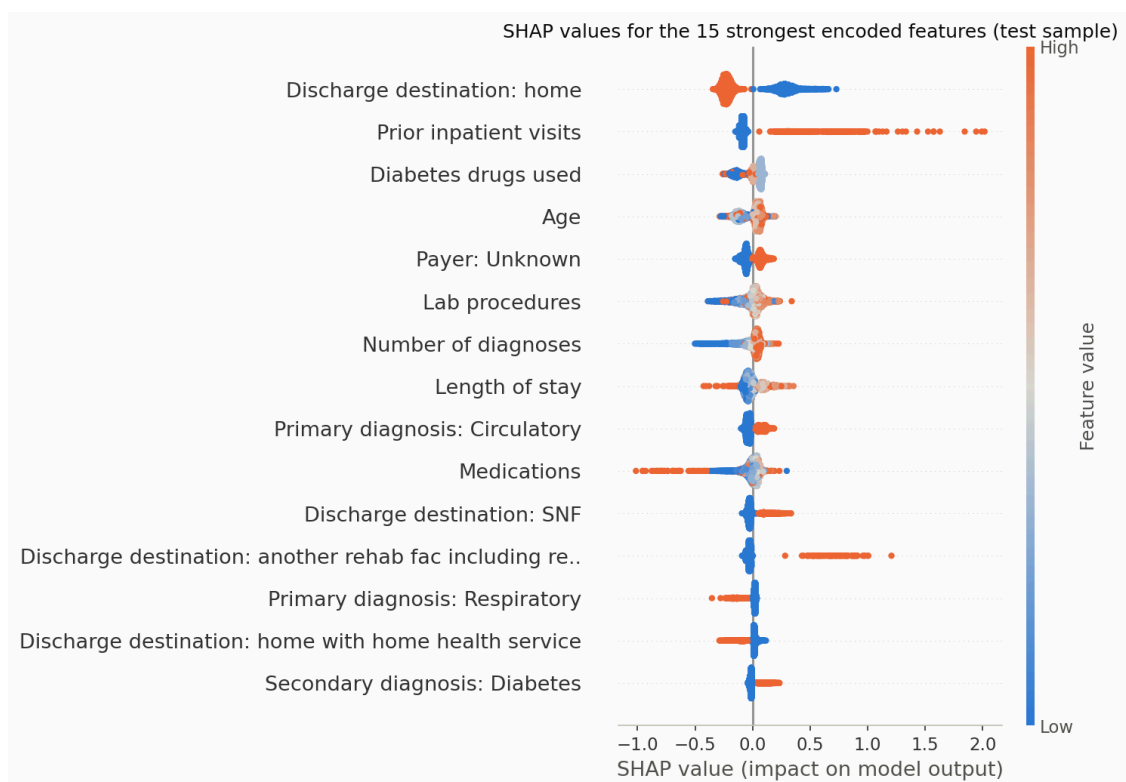


Figure 6.6 SHAP values for the 15 strongest encoded features on a test sample. Orange means a high feature value, blue a low one. Points to the right push risk up.

6.7 Risk tiers

Every patient gets an out-of-sample score: training patients get their out-of-fold prediction and test patients get the final model's prediction. No patient is scored by a model that trained on them, so the tiers used in Power BI are honest. Tiers are cut at the 50th and 80th percentiles of the scores, and each patient also gets the three features pushing their risk up the most.

Table 6.6 Risk tiers. Cut-offs: Low below 0.075, High from 0.115.

risk_tier	patients	mean_predicted	observed_rate	readmissions	share_of_readmissions
Low	34,985	5.3%	5.5%	1,924	30.7%
Medium	20,991	9.2%	9.3%	1,946	31.0%
High	13,994	17.4%	17.2%	2,407	38.3%

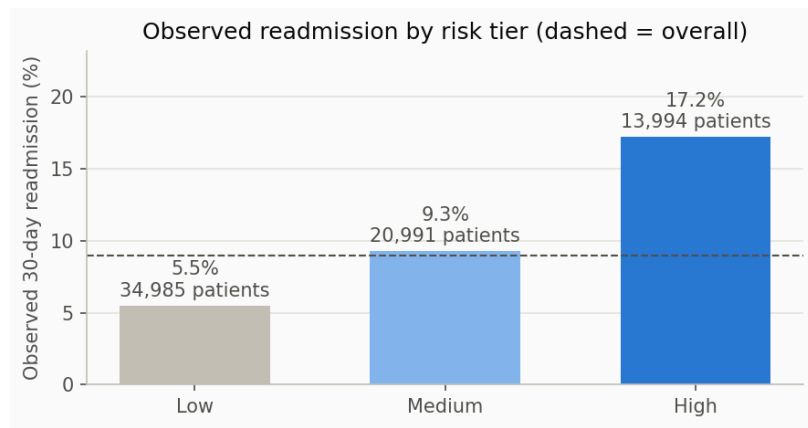


Figure 6.7 Observed readmission by risk tier. Dashed line is the overall rate.

The high tier is 20% of patients and holds 38% of all readmissions. Its observed rate (17.2%) is three times the low tier (5.5%), and the predicted and observed rates match in every tier.

7 From analysis to dashboard

Notebook 07. Preparing the results for the Power BI dashboard.

Notebook 07 only exports tables. Relationships, DAX measures, formatting and design are all built inside Power BI. The export follows a star schema: one fact table with one row per patient (including the model's risk score, tier and top three drivers), two long-format fact tables for medications and risk drivers, seven dimension tables with sort orders, and the statistical and model results from Sections 5 and 6.

Table 7.1 Tables exported to outputs/powerbi.

Table	Rows	Content
fact_encounter	69,970	One row per patient, all measures, risk score, tier, drivers
fact_medication	83,273	One row per patient per drug used
fact_driver	209,906	Top 3 risk drivers per patient, long format
dim_ (7 tables)	3 to 69,970	Patient, diagnosis, discharge, admission type and source, specialty, risk tier
stat_ (5 tables)	3 to 37	Odds ratios, HbA1c by diagnosis, methods, PSM balance, what-if scenario
model_ (8 tables)	2 to 6,486	Comparison, curves, calibration, confusion matrix, SHAP, deciles, tiers

Before the files are written, ten integrity checks run: patients are unique, every key in the fact table exists in its dimension, and the readmission rate matches notebook 03. All ten pass, so no visual in Power BI will show a blank category.

7.1 What-if: testing more patients

The dashboard has a what-if page for HbA1c testing. It uses the **adjusted** odds ratios by diagnosis from Section 5, not raw rates, so the scenario does not repeat the confounding removed there. For each diagnosis, the implied readmission rate if untested patients were tested is $p \times OR / (1 - p + p \times OR)$, where p is the current rate among untested patients. Only the three diagnoses with a significant effect count towards the total.

Code 7.1 What-if scenario for HbA1c testing.

07_powerbi_export.ipynb, cell 9

```
g = f.assign(diag=f.diag_1_group.replace({'Missing': 'Other'})).query('alc_tested == 0').groupby('diag').agg(untested_patients=('readmit_30', 'size'), rate_untested=('readmit_30', 'mean'))
sc = g.join(OUT['stat_alc_by_diagnosis'].set_index('label')[['OR', 'ci_low', 'ci_high', 'significant']])
sc['rate_if_tested'] = sc.rate_untested*sc.OR/(1 - sc.rate_untested + sc.rate_untested*sc.OR)
sc['avoided_if_all_tested'] = sc.untested_patients*(sc.rate_untested - sc.rate_if_tested)
```

Table 7.2 Projected readmissions avoided if every untested patient in the three significant diagnoses were tested.

diag_group	untested_patients	rate_untested	OR	rate_if_tested	avoided_if_all_tested
Diabetes	3,687	10.2%	0.814	8.4%	64
Injury	4,115	11.3%	0.612	7.2%	167
Respiratory	7,622	7.7%	0.761	6.0%	132

Total: about 363 fewer readmissions across 15,424 untested patients. This assumes the adjusted association is causal, which the data cannot confirm.

The eight dashboard pages cover an executive summary, patients and stays, readmission drivers, diagnosis and diabetes care, medication, the statistical and model results, risk and what-if analysis with adjustable threshold and cost assumptions, and a patient profile reached by drill-through from the highest-risk patient list.

8 Conclusions and limitations

What the analysis found, where it stops, and what would come next.

8.1 Findings

1. **Prior hospital use is the strongest risk factor.** Patients with three or more inpatient stays in the previous year are readmitted at 26.5% (relative risk 3.26, adjusted OR 3.61). This held in every method used.
2. **Discharge destination is the second.** Transfers to another facility come back at 15.2% vs 7.0% for patients sent home (adjusted OR 2.17), and it is the top driver in the model.
3. **HbA1c is rarely measured**, for 18.4% of patients and only 35.9% of those admitted for diabetes. Across all patients testing is not linked to readmission after adjustment (adjusted OR 0.96, matched OR 0.99), but it is linked to lower odds in injury, respiratory and diabetes admissions (OR 0.61 to 0.81), and the interaction is significant ($p = 0.0067$).
4. **Readmission can be predicted, but only moderately.** XGBoost reaches a test ROC-AUC of 0.659 and PR-AUC of 0.183, with good calibration. The top 20% of patients by risk hold 38% of readmissions.

8.2 What a hospital could do with this

- Use the risk score to order a follow-up list. Starting from the top 20% reaches almost twice as many readmissions as a random list of the same size.
- Give patients with prior inpatient stays and patients being transferred to another facility a structured hand-over and an early follow-up call, since both groups carry high risk and are easy to identify at discharge.
- Look at HbA1c testing for injury, respiratory and diabetes admissions. The test is cheap and rarely done, and those are the groups where it is linked to fewer readmissions. A prospective check would be needed before calling it a cause.

8.3 Limitations

- **The data is old.** It covers 1999 to 2008. Care practice, coding (ICD-9 has since been replaced by ICD-10) and readmission rates have all changed.
- **No dates.** The first encounter is taken as the lowest encounter ID, which is an approximation. Later encounters are excluded, so the model is not validated on repeat admissions.
- **Observational design.** The HbA1c results are adjusted associations. Unmeasured factors such as illness severity, lab values or the care team could still explain them.
- **Missing context.** Specialty is unknown for half the patients and payer for 40%. Social and follow-up information is absent, which caps what any model can do.

- **Assumptions.** The 10 to 1 cost ratio and the what-if projection are assumptions for illustration, not measured values.
- **No external validation.** The model is tested on a held-out sample of the same data, not on another hospital or a later period.

8.4 Next steps

- Check the model for fairness: calibration and sensitivity by race, gender and age group, not only the raw rates in Q12.
- Test the model on a time-based split or another hospital system if data with dates becomes available.
- Try adding repeat encounters with patient-level grouping in cross-validation, which would use the 30% of encounters dropped here.

A Reproducing the analysis

All notebooks run on the free Google Colab CPU and read and write inside one Drive folder. Run them in order, 01 to 07. Notebook 06 takes about 5 to 10 minutes, mostly for the XGBoost search. Every notebook skips steps whose output already exists or overwrites it, so any one can be rerun on its own. To run outside Colab, change the `PROJ` path in the first cell of each notebook and remove the `drive.mount` line.

```
Diabetes Readmission Project/
├── data/
│   ├── raw/           diabetic_data.csv, IDS_mapping.csv
│   ├── parquet/       raw data and 3 lookup tables as Parquet
│   └── processed/     encounters_clean, encounters_enriched, patient_risk_scores, audit_log
├── notebooks/        01_download_data to 07_powerbi_export
├── outputs/
│   ├── figures/       20 PNG charts used in this report
│   ├── tables/        statistical and model result tables (04_, 05_, 06_)
│   ├── sql_results/   q01 to q13, one CSV per SQL question
│   ├── models/        readmission_model.joblib (fitted XGBoost pipeline)
│   └── powerbi/       23 CSV tables for the dashboard
├── Diabetes Readmission Project.pbix
└── Diabetes Readmission Project.pdf   (dashboard export)
```

Main libraries: DuckDB, pandas, NumPy, SciPy, statsmodels, scikit-learn, XGBoost, SHAP, Matplotlib. Random seed 42 throughout.

References

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